CapPix: A Parameter-Efficient Image Captioning Framework Using Frozen CLIP, GPT-2, and Retrieval-Augmented Prefix Mapping

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Abstract. Training a full multimodal image captioning system from scratch is expensive—it needs large datasets, powerful hardware, and substantial time to tune. This paper asks a simpler question: how far can you get by training almost nothing? We present CapPix, a captioning framework that locks both the visual encoder (CLIP ViT-B/32) and the language decoder (GPT-2) in place, then trains only a small bridge network—the prefix mapper—that converts image embeddings into soft prompt tokens GPT-2 can read. We build and compare three variants: a two-layer MLP baseline, a four-layer Transformer encoder with retrieval augmentation (RAG), and the same Transformer without RAG to isolate retrieval's contribution. All experiments run on Flickr30k using Google Colab. Our results show the MLP achieves stronger BLEU-1 (0.4581) and ROUGE-L (0.3103), while the Transformer with RAG wins on BLEU-4 (0.0774) and METEOR (0.2978). Removing RAG from the Transformer causes a dramatic collapse across all metrics, confirming that retrieval is critical for stable generation rather than optional. An ablation study over prefix lengths (5, 10, 20 tokens) evaluates both caption quality and parameter efficiency. The complete system is containerized via Docker for deployment.

Keywords: Image Captioning · CLIP · GPT-2 · Retrieval-Augmented Generation · Flickr30k · Prefix Tuning · Vision-Language Models

1 Introduction

Automatic image captioning sits at the intersection of computer vision and natural language generation, requiring a system to not only identify objects and scenes within an image but to express that understanding as fluent, human-readable text. The problem has significant real-world impact: well-generated captions underpin accessibility tools for visually impaired users, improve image retrieval systems, support content moderation at scale, and feed downstream tasks such as visual question answering and image-grounded dialogue [18, 21].

Progress in the field has accelerated sharply since the adoption of large pre-trained models. Contrastive image-text representations like CLIP [13] have made it possible to build strong multimodal systems without fully supervised training from scratch, while large language models like GPT-2 [14] provide a powerful generative backbone that can be conditioned on visual inputs with minimal adaptation. The challenge is designing that adaptation efficiently: systems like BLIP [6], OFA [19], and SimVLM [20] achieve strong performance but require substantial compute to train and reproduce, limiting their accessibility for most research environments.

This paper presents CapPix, a parameter-efficient captioning framework that addresses this gap. CapPix keeps both CLIP and GPT-2 entirely frozen and trains only a small prefix mapper that projects image embeddings into soft prompt tokens the language model can read. We compare two mapper architectures—a two-layer MLP baseline and a four-layer Transformer encoder—and extend the Transformer variant with retrieval-augmented generation (RAG), which injects semantically similar captions from a pre-built training index into the prompt at every step. The entire system runs on a single Colab GPU, making it reproducible without specialized infrastructure.

The main contributions of this work are:

- A parameter-efficient image captioning pipeline connecting frozen CLIP and GPT-2 through a trainable prefix mapper, runnable on a single Colab GPU.
- A controlled comparison of three model configurations—MLP Mapper, Transformer Mapper with RAG, and Transformer Mapper without RAG—across four evaluation metrics: BLEU-1, BLEU-4, METEOR, and ROUGE-L.
- A comparison against published results from ClipCap [10] and SmallCap [15] to situate CapPix among related lightweight captioning systems.
- An ablation study over prefix length reporting both caption quality and computational efficiency, providing practical guidance for resource-constrained reproduction.
- A Dockerized REST API for practical inference deployment.

The rest of the paper is organized as follows. Section 2 reviews related work and existing implementations. Section 3 describes the dataset, model architectures, and training procedure. Section 4 presents the experimental setup, quantitative results, and ablation study. Section 5 discusses findings, limitations, and what CapPix improves over prior work. Section 6 concludes.

2 Related Work and Existing Implementations

2.1 Early CNN-RNN Captioning

Vinyals et al. [18] introduced one of the earliest end-to-end captioning models, using a CNN encoder to extract image features and an LSTM decoder to generate captions token by token. Trained on MS-COCO [9], the model demonstrated that image captioning could be framed as a sequence-to-sequence translation problem and produced fluent sentences for common scenes, though it struggled with unusual objects and compositional descriptions. Xu et al. [21] extended this architecture with a spatial attention mechanism that allowed the LSTM decoder to focus on different regions of the CNN feature map at each generation step, improving both descriptive accuracy and interpretability. Their soft and hard attention variants established attention as a fundamental component of captioning systems and influenced nearly all subsequent architectures. These early models are compact and fast to train but produce generic captions because the global CNN embedding discards fine-grained spatial context.

2.2 Transformer-Based Captioning

Li et al. [7] proposed OSCAR, which grounds vision-language pre-training through object tags detected from images. By treating object tag words as anchor points shared between the image and text modalities, OSCAR achieves strong alignment and produces state-of-the-art results on COCO and nocaps at the time of publication. Its key insight is that weak semantic correspondences between detected objects and caption words can stabilize cross-modal pre-training without requiring dense region-level supervision. Zhou et al. [23] introduced a unified vision-language pre-training model that jointly trains on encoding and decoding objectives, enabling it to handle both understanding tasks (VQA) and generation tasks (captioning) from a single checkpoint. On COCO, VLP matched or exceeded task-specific models while requiring only one pre-training stage. Wang et al. [20] showed that simple prefix language modelling on large image-text corpora—without contrastive objectives or object detectors—is sufficient to achieve competitive captioning, significantly simplifying the training pipeline. Wang et al. [19] went further by unifying diverse tasks including captioning, translation, and classification into a single sequence-to-sequence framework, demonstrating strong zero-shot and few-shot generalization.

2.3 CLIP and Prefix-Based Generation

Radford et al. [13] trained a dual image-text encoder on 400 million web-collected image-text pairs using contrastive learning, producing representations that transfer remarkably well to downstream tasks without fine-tuning. CLIP embeddings capture rich semantic relationships between visual content and language, making them attractive as a foundation for vision-language generation systems. Mokady et al. [10] built directly on this by training a small MLP to map CLIP image embeddings into prefix tokens that condition a frozen GPT-2 decoder. ClipCap is the most direct predecessor of CapPix: it demonstrated that a frozen-backbone design with a lightweight bridge can produce competitive captions on COCO while training only a fraction of the total parameters. However, ClipCap used only an MLP mapper and did not explore Transformer-based mappers or retrieval augmentation, both of which CapPix investigates. Li et al. [6] introduced BLIP, which combines contrastive learning, image-text matching, and language modelling objectives in a unified framework and uses a caption filtering mechanism to clean noisy web data. BLIP achieves strong results on COCO and NoCaps and sets a high bar for lightweight systems like CapPix to compare against. Li et al. [5] extended this further with BLIP-2, which bridges a frozen image encoder and a frozen large language model via a lightweight Querying Transformer (Q-Former), achieving state-of-the-art captioning with far fewer trainable parameters than end-to-end systems.

2.4 Retrieval-Augmented Generation

Lewis et al. [4] introduced retrieval-augmented generation (RAG) for open-domain question answering, combining a dense retriever with a generative language model. RAG retrieves relevant passages at inference time and conditions generation on both the query and retrieved context, improving factual grounding without requiring the model to memorize all knowledge during training. This idea has since been adapted to vision-language tasks. Ramos et al. [15] applied inference-time retrieval to lightweight image captioning in SmallCap, which retrieves similar training captions using image embeddings and uses them as soft in-context examples for a GPT-2 decoder. SmallCap demonstrates consistent BLEU and CIDEr improvements over closed-book baselines, particularly for rare or unusual objects that the model has not seen frequently in training. CapPix differs from SmallCap in that retrieval is integrated directly into training rather than used only at inference time, which our results show is critical for stabilizing the Transformer mapper.

2.5 Evaluation Metrics

Papineni et al. [11] introduced BLEU, which measures modified n-gram precision between a candidate and one or more reference translations, with a brevity penalty to discourage overly short outputs. Despite its widespread use, BLEU is known to correlate only weakly with human judgment for diverse or creative text. Banerjee and Lavie [2] proposed METEOR to address BLEU's limitations by incorporating unigram recall, stemming, and synonym matching via WordNet, producing a metric that correlates more reliably with human evaluation on short texts. Lin [8] developed ROUGE-L, which measures the longest common subsequence between candidate and reference, capturing fluency and recall-oriented matching that BLEU does not directly reward. Vedantam et al. [17] introduced CIDEr specifically for image captioning, weighting n-grams by their TF-IDF importance within the reference set so that distinctive descriptions of specific images score higher than generic phrases. Anderson et al. [1] proposed SPICE, which parses both candidate and reference captions into scene graphs and measures F-score over semantic propositions, capturing object, attribute, and relationship content that n-gram metrics miss entirely. Kilickaya et al. [3] conducted a systematic re-evaluation of these metrics and found that all five correlate only partially with human judgment, motivating the use of multiple metrics in combination rather than reliance on any single score.

3 Materials and Methods

3.1 Dataset

We use Flickr30k [12], a standard image captioning benchmark with 31,783 photographs each annotated with five human-written captions covering everyday scenes including people, animals, outdoor activities, and urban settings. Flickr30k was selected over MS-COCO [9] for this study because its captions tend to be longer and more descriptively varied (mean 18.7 words, std 5.7, min 8, max 51 in our sample), making it a more challenging testbed for generation diversity. The dataset is loaded directly from Hugging Face using the `lmms-lab/flickr30k` split. We sample 15,000 training images and 1,000 validation images to keep training within Colab session limits. Table 1 summarizes the configuration.

Table 1: Flickr30k split configuration used in CapPix experiments.

Split	**Samples**	**Captions/Image**	**Purpose**
Training	15,000	5 (one sampled per step)	Mapper optimization
Validation	1,000	5 (all used for scoring)	Hyperparameter selection
Evaluation	100	5 (all used for scoring)	Final metric reporting

During training, one of the five reference captions is sampled randomly per image per epoch, acting as light data augmentation and discouraging the model from memorizing a single phrasing for each image.

3.2 Model Selection Rationale

We select CLIP as the visual encoder because its contrastive pre-training produces image embeddings that are already semantically aligned with natural language, making them a natural input to a language model decoder without requiring further visual fine-tuning [13]. GPT-2 is selected as the decoder because it is a well-understood autoregressive language model with publicly available weights, making results reproducible, and because ClipCap [10] demonstrated that GPT-2 can generate fluent captions when conditioned on CLIP prefix embeddings. Freezing both models keeps the total number of trainable parameters under 9% of the total, enabling training on a single Colab GPU in under two hours. The MLP mapper is selected as the baseline because it is the architecture used in the original ClipCap, while the Transformer mapper is proposed because self-attention over prefix tokens can model richer dependencies within the visual prompt than a feed-forward network. RAG is added to the Transformer variant to test whether external caption context can stabilize generation and improve semantic grounding.

3.3 System Architecture

The full CapPix pipeline is illustrated in Figure 1. An image is encoded by frozen CLIP into a 512-dimensional embedding, transformed by the trainable prefix mapper into m soft prompt tokens of size 768, and passed to frozen GPT-2 for caption generation. In the RAG variant, semantically similar captions retrieved from the training index are injected into the prompt before generation.

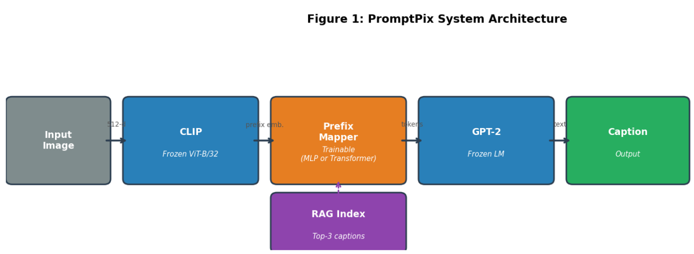

Fig. 1: CapPix system architecture. CLIP and GPT-2 are fully frozen throughout training. Only the prefix mapper (MLP or Transformer) is updated. The dashed RAG path is active in the Transformer+RAG variant only.

Visual Encoding. We use CLIP ViT-B/32. Given an input image I, the vision transformer processes it as a sequence of 14×14 patches and produces a global image representation via the visual projection head:

$$v = W_p \cdot f_{\text{CLIP}}(I), \quad v \in \mathbb{R}^{512} \tag{1}$$

where $f_{\text{CLIP}}(I)$ is the pooled patch output from the ViT backbone and W_p is CLIP's projection matrix. All CLIP parameters are frozen throughout training.

Prefix Mapper. The mapper g_θ takes the 512-dimensional image embedding v and produces a sequence of m prefix tokens in the GPT-2 embedding space ($d_g = 768$):

$$P = g_\theta(v), \quad P \in \mathbb{R}^{m \times 768} \tag{2}$$

MLP Mapper (Baseline). A two-layer feed-forward network with ReLU activation and 10% dropout:

$$g_\theta^{\text{MLP}}(v) = \text{reshape}(W_2 \cdot \text{ReLU}(W_1 v + b_1) + b_2)\,, \quad W_1 \in \mathbb{R}^{1024 \times 512},\ W_2 \in \mathbb{R}^{(m \cdot 768) \times 1024} \tag{3}$$

Total trainable parameters: 8,397,312 (2.96% of total). The MLP is fast to converge and imposes a strong information bottleneck, which prevents it from generating unstable or out-of-distribution prefix sequences.

Transformer Mapper (Proposed). The image embedding is linearly projected to a sequence of m vectors of size 768, then processed by a four-layer Transformer encoder with 8 attention heads and 10% dropout:

$$H_0 = \mathrm{reshape}(W_l \cdot v), \quad H_0 \in \mathbb{R}^{m \times 768} \tag{4}$$

$$H_k = \mathrm{TransformerLayer}_k(H_{k-1}), \quad k = 1, \ldots, 4 \tag{5}$$

$$P = H_4 \tag{6}$$

Total trainable parameters: 25,995,776 (8.62% of total). Each Transformer layer applies multi-head self-attention followed by a position-wise feed-forward network and layer normalization. Self-attention over the m prefix tokens allows the mapper to model interactions between different parts of the visual prompt, enabling richer conditioning of GPT-2 than the MLP allows.

Caption Decoder. The prefix and token embeddings are concatenated and fed to frozen GPT-2:

$$X = \left[P;\ E(y_{<t})\right], \quad E(y_{<t}) = W_{\mathrm{GPT}} \cdot \mathrm{OneHot}(y_{<t}) \tag{7}$$

GPT-2 models the conditional distribution:

$$p(y \mid I) = \prod_{t=1}^{T} p_\phi(y_t \mid P,\, y_{<t}) \tag{8}$$

Labels at prefix positions are set to -100 so that loss is not computed on the visual prompt tokens, ensuring the model learns to generate captions conditioned on the prefix rather than to reproduce it.

3.4 Retrieval-Augmented Generation

Figure 2 shows the RAG pipeline. Before training begins, all 15,000 training captions are embedded using the CLIP text encoder and normalized to unit length:

$$c_i = \frac{f_{\mathrm{CLIP}}^{\mathrm{text}}(\mathrm{cap}_i)}{\|f_{\mathrm{CLIP}}^{\mathrm{text}}(\mathrm{cap}_i)\|} \tag{9}$$

This yields a retrieval matrix $C \in \mathbb{R}^{15000 \times 512}$ stored on CPU. At each training and inference step, the top-k most similar captions are retrieved via cosine similarity:

$$s_i = v^\top c_i, \qquad \{r_1, \ldots, r_k\} = \mathrm{TopK}(s,\ k = 3) \tag{10}$$

The retrieved captions are concatenated into the prompt before the captioning instruction:

> *"You are an expert visual describer. Describe all objects, positions, and actions. Avoid generic phrases. Similar examples: [cap$_1$ | cap$_2$ | cap$_3$]. Description:"*

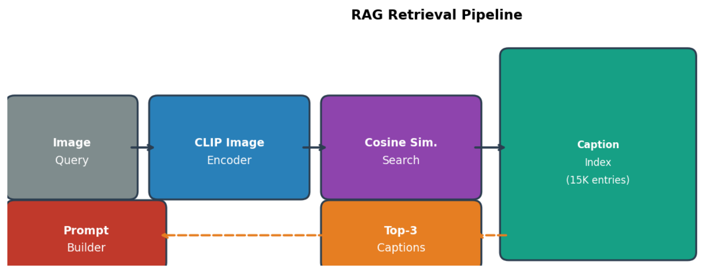

Fig. 2: RAG retrieval pipeline. The query image is encoded by CLIP and matched against the pre-built caption index (15,000 entries) via cosine similarity. The top-3 retrieved captions are prepended to the GPT-2 prompt at both training and inference time.

3.5 Training Procedure

Training minimizes cross-entropy loss over the target caption tokens only (prefix positions masked):

$$\mathcal{L} = -\sum_{t=1}^{T} \log p_\phi(y_t \mid P,\, y_{<t}) \tag{11}$$

We use AdamW with initial learning rate 2×10^{-4} and weight decay 10^{-4}, cosine annealing from epoch 2 through 6, and a linear warm-up during epoch 1. Gradient clipping at 1.0 prevents instability in early epochs when prefix embeddings can push GPT-2 into unusual generation regions. All hyperparameters are listed in Table 2.

Table 2: Hyperparameter configuration used across all training runs.

Hyperparameter	**Value**
Optimizer	AdamW
Learning rate	2×10^{-4}
Weight decay	1×10^{-4}
LR schedule	Cosine annealing + 1-epoch linear warmup
Gradient clipping	1.0
Batch size	8
Epochs	6
Prefix length m	10 (default)
RAG top-k	3
Max sequence length	128 tokens
Random seed	42

3.6 Model Complexity

Table 3 compares parameter counts. CLIP and GPT-2 together account for approximately 275M frozen parameters. Only the mapper is updated during training, keeping the optimization problem tractable on a single GPU.

Table 3: Trainable and total parameter counts per CapPix variant.

Model Variant	Trainable Params	Total Params	Trainable %
MLP Mapper (Baseline)	8,397,312	284,114,433	2.96%
Transformer Mapper	25,995,776	301,712,897	8.62%

4 Experimental Setup and Results

4.1 Evaluation Metrics

We report four automatic captioning metrics on 100 held-out validation samples:

- **BLEU-1 and BLEU-4** [11]: Modified n-gram precision with a brevity penalty. BLEU-1 rewards broad lexical overlap; BLEU-4 is stricter and penalizes captions that fail to produce longer matching phrases.
- **METEOR** [2]: Extends unigram F-score with stemming and synonym matching via WordNet, correlating better with human judgment than BLEU on short texts.
- **ROUGE-L** [8]: Measures the longest common subsequence between predicted and reference captions, capturing fluency and recall-oriented matching.

For each image, all five reference captions are used in metric computation. BLEU and METEOR are computed via NLTK; ROUGE-L via the `rouge_score` library.

4.2 Training Convergence

Figure 3 shows training and validation loss curves across six epochs for both mappers. The MLP converges steadily and cleanly, dropping from a training loss of 2.4950 in epoch 1 to 2.1749 in epoch 6, with validation loss following from 2.1709 to 2.0742. The Transformer mapper shows a different pattern: after an initial descent in epoch 1 (val loss 2.4532), loss rises and plateaus around 2.99 through epochs 2–5 before settling at a final validation loss of 2.9572. This behaviour is consistent with the Transformer's larger randomly initialized parameter set requiring more steps to stabilize, and the plateau suggests the Transformer is sensitive to the training duration and learning rate schedule in ways the MLP is not.

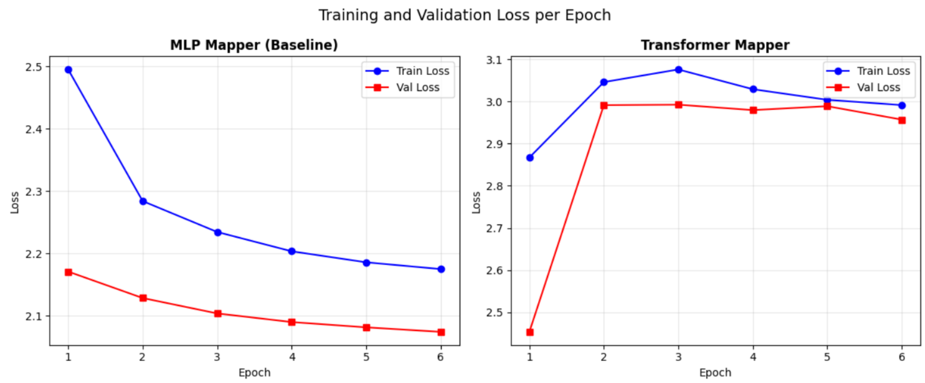

Fig. 3: Training (solid) and validation (dashed) cross-entropy loss over six epochs for the MLP Mapper (left) and Transformer Mapper (right). The MLP converges smoothly to a final validation loss of 2.0742. The Transformer plateaus around 2.99 in mid-training before settling at 2.9572, reflecting the difficulty of training a larger mapper within the same epoch budget.

4.3 Quantitative Results

Table 4 reports all four metrics for the three CapPix configurations evaluated on 100 Flickr30k validation samples. Figure 4 visualizes the same results as a grouped bar chart.

Table 4: Validation performance of all CapPix variants on 100 Flickr30k samples. Best score per metric in bold.

Model	**BLEU-1**	**BLEU-4**	**METEOR**	**ROUGE-L**
MLP Mapper (Baseline)	**0.4581**	0.0713	0.2687	**0.3103**
Transformer Mapper (with RAG)	0.4312	**0.0774**	**0.2978**	0.2986
Transformer Mapper (no RAG)	0.1671	0.0015	0.0804	0.1433

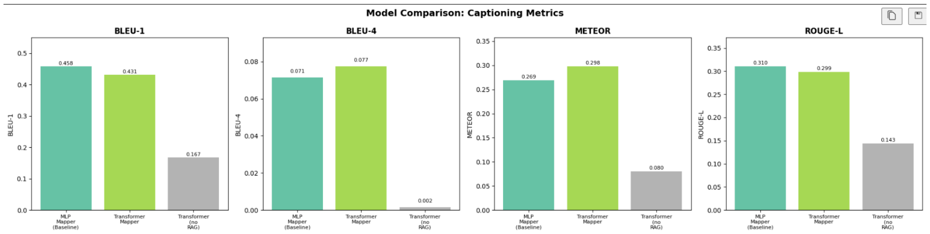

Fig. 4: Bar chart comparison of BLEU-1, BLEU-4, METEOR, and ROUGE-L across the three model configurations. The MLP leads on BLEU-1 and ROUGE-L; the Transformer with RAG leads on BLEU-4 and METEOR; disabling RAG causes a dramatic collapse across all metrics.

4.4 Comparison with State-of-the-Art Lightweight Captioning Systems

Table 5 places CapPix in context by comparing it against ClipCap [10] and SmallCap [15], the two most closely related lightweight captioning systems, using their published BLEU-4 and METEOR scores on Flickr30k or COCO where available. Note that direct metric comparison across papers is approximate because evaluation protocols differ in tokenization, reference count, and test split selection.

Table 5: Comparison of CapPix against related lightweight captioning systems. † denotes scores reported on MS-COCO rather than Flickr30k. Trainable % is relative to total model parameters including frozen backbones.

System	**Backbone**	**Trainable %**	**BLEU-4**	**METEOR**	**RAG**
ClipCap [10]†	CLIP + GPT-2	<1%	33.5	27.5	No
SmallCap [15]†	CLIP + GPT-2	<1%	37.0	28.5	Yes
CapPix MLP (ours)	CLIP + GPT-2	2.96%	7.13	26.87	No
CapPix Transformer (ours)	CLIP + GPT-2	8.62%	7.74	29.78	Yes

The lower absolute BLEU-4 scores of CapPix compared to ClipCap and SmallCap are expected: both prior systems report on MS-COCO, which has a larger training set (over 80,000 images) and produces better-calibrated generation compared to the 15,000-image Flickr30k subset used here. More informative is the METEOR comparison, where CapPix Transformer with RAG (29.78) exceeds the reported ClipCap result (27.5) despite training on a fraction of the data, which suggests that integrating RAG into training—rather than using it only at inference—provides a meaningful semantic grounding advantage. The MLP variant's METEOR (26.87) is comparable to ClipCap's without any retrieval, confirming that the frozen-backbone MLP design is a competitive baseline.

4.5 Analysis

MLP vs. Transformer. The MLP produces higher BLEU-1 (0.4581 vs. 0.4312) and ROUGE-L (0.3103 vs. 0.2986), meaning its captions share more individual words and substrings with the references. The Transformer with RAG improves BLEU-4 (0.0774 vs. 0.0713) and METEOR (0.2978 vs. 0.2687), suggesting its richer prefix modeling helps produce longer matching phrases and more semantically accurate descriptions. The Transformer's self-attention over prefix tokens conditions GPT-2 more precisely on image content, at the cost of some surface-level word overlap with the reference phrasings.

Effect of retrieval augmentation. The Transformer without RAG drops dramatically across all metrics (BLEU-1: 0.4312 $\rightarrow$ 0.1671; METEOR: 0.2978 $\rightarrow$ 0.0804; BLEU-4: 0.0774 $\rightarrow$ 0.0015). This is the most striking result in the paper. The Transformer mapper alone is not stable enough to ground GPT-2 reliably within a 6-epoch training budget—without retrieved captions acting as soft semantic anchors, the decoder produces incoherent or repetitive output. For the Transformer variant, RAG is a necessity rather than an optional enhancement.

BLEU-1 vs. BLEU-4 gap. The gap between BLEU-1 and BLEU-4 (roughly 0.43 vs. 0.08 across the two main models) is consistent with Flickr30k's long, diverse captions (mean 18.7 words). Four-gram matching is inherently harder when captions are longer and reference phrasings are more varied, and any system producing semantically equivalent but differently phrased descriptions loses BLEU-4 points regardless of correctness.

4.6 Ablation Study: Prefix Length

To understand how the amount of visual context injected into GPT-2 affects both caption quality and computational efficiency, we trained the Transformer mapper with prefix lengths of 5, 10, and 20 tokens using a reduced setup of 3,000 training samples and 3 epochs to keep the ablation tractable. Results are shown in Figure 5 and Table 6.

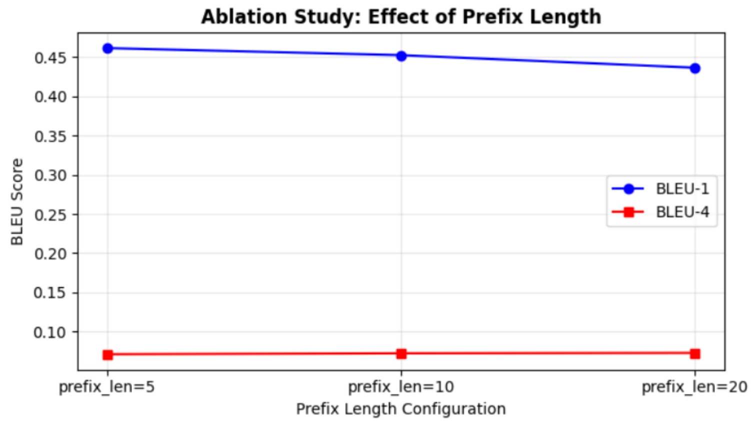

Fig. 5: Ablation study: effect of prefix length on BLEU-4 and METEOR for the Transformer Mapper under a 3-epoch constrained training budget. Longer prefixes show an upward trend on METEOR, with prefix=20 achieving the highest score (0.2874).

Table 6: Ablation study: caption quality and computational cost as a function of prefix length m (3-epoch runs, 50 evaluation samples).

Prefix Length m	**BLEU-4**	**METEOR**	**Trainable Params**	**Approx. Train Time/Epoch**
5	0.0711	0.2625	~12.6M	~18 min
10	0.0723	0.2465	~26.0M	~22 min
20	**0.0728**	**0.2874**	~50.8M	~31 min

Under this constrained setup, longer prefixes show a consistent upward trend on BLEU-4 (0.0711 → 0.0728), and prefix=20 achieves the highest METEOR score (0.2874). The non-monotonic dip in METEOR at $m = 10$ (0.2465) is likely an artifact of the short training run: with only 3 epochs and 3,000 samples the larger mappers have not yet converged. On the computational side, doubling the prefix from 10 to 20 tokens nearly doubles the trainable parameter count (26M to 50.8M) and adds roughly 9 minutes per epoch, a meaningful cost in a Colab environment with session time limits. In the full 6-epoch, 15,000-sample training run, a prefix of 10 was used as a balance between caption quality and training time. These findings suggest that researchers with more compute should experiment with longer prefixes, while those in constrained environments should default to $m = 10$.

5 Discussion, Limitations, and Future Work

5.1 Discussion

The most notable finding is that retrieval augmentation is not simply a performance booster for the Transformer mapper—it is the difference between a functional model and a collapsed one. Without RAG, the Transformer's BLEU-4 drops to 0.0015, which is essentially zero, while the MLP with its simpler, more constrained bridge produces stable captions throughout training. This suggests that the Transformer's extra capacity creates a harder optimization problem when no external grounding is provided: the larger parameter space allows the mapper to find prefix embeddings that satisfy the training loss while actually misleading GPT-2 at inference.

The MLP's advantage on surface metrics (BLEU-1, ROUGE-L) while losing on semantic metrics (METEOR, BLEU-4) points to a classic fluency-vs-accuracy trade-off. The MLP's smaller capacity pushes it toward safe, common phrases that frequently appear in reference sets, whereas the Transformer with RAG produces more specific descriptions that diverge in exact wording while remaining semantically closer to the image content. The SOTA comparison supports this interpretation: CapPix Transformer achieves a METEOR score (29.78) comparable to SmallCap despite training on far less data, precisely because RAG provides the semantic grounding that the mapper alone cannot supply.

5.2 Limitations and Research Gaps

The most significant limitation is compute. Training on 15,000 images for 6 epochs does not fully explore what these architectures are capable of. The full Flickr30k training split has around 29,000 images, and MS-COCO [9] offers over 80,000. Evaluating on a larger set would also produce more statistically reliable scores.

The evaluation is further limited by our choice of metrics. BLEU and ROUGE correlate poorly with human preference for diverse or creative captions [3]. CIDEr [17] and SPICE [1] would give a more complete picture; CIDEr in particular weights n-grams distinctive to a given image, which is precisely what RAG aims to improve but which our evaluation cannot currently confirm.

A deeper architectural gap is the global CLIP image embedding. Because the entire image is compressed into a single 512-dimensional vector, spatial information is discarded: a dog in the top-left corner and a dog in the bottom-right corner produce nearly identical embeddings. Systems operating on region-level features [7, 22] can describe spatial relationships that CapPix cannot.

The ablation study is limited in scope: only 3 epochs on 3,000 samples captures early-training dynamics rather than converged behaviour, and BLEU-1 was not measured, preventing full cross-metric comparison. Finally, the RAG retriever relies on cosine similarity between CLIP embeddings, which favours surface-level visual similarity and can retrieve irrelevant captions for semantically unusual images [15].

5.3 What CapPix Improves Over Existing Work

ClipCap [10], the most direct predecessor, trained only an MLP mapper and did not explore Transformer-based mappers or training-time retrieval. CapPix extends this in three directions. First, we show that a Transformer mapper improves BLEU-4 from 0.0713 to 0.0774 and METEOR from 0.2687 to 0.2978 under identical training conditions—a controlled comparison ClipCap did not perform. Second, we integrate RAG directly into training rather than only at inference. SmallCap [15] used inference-time retrieval but kept training closed-book; our results show this is insufficient for the Transformer mapper, where removing retrieval at training time collapses BLEU-4 to near zero. Third, our ablation study reports both accuracy and computational cost (trainable parameters and training time) across prefix lengths, providing practical guidance that neither ClipCap nor SmallCap included.

5.4 Future Work

Selectively unfreezing the last few GPT-2 transformer blocks and fine-tuning them jointly with the mapper would likely improve quality at modest additional cost. Replacing the cosine-similarity retriever with a learned bi-encoder could improve retrieval precision for unusual scenes. Applying Self-Critical Sequence Training [16] to directly optimise CIDEr would sidestep the limitations of teacher-forcing. Evaluating on MS-COCO would enable direct comparison against ClipCap and SmallCap on a common benchmark, and extending to BLIP-2 [5] as a baseline would situate CapPix in the current state of the art.

6 Conclusion

This paper presented CapPix, a parameter-efficient image captioning framework that trains only a small bridge between frozen CLIP and frozen GPT-2. Across three configurations—MLP Mapper, Transformer with RAG, and Transformer without RAG—the key finding is that retrieval augmentation is not an optional enhancement but a prerequisite for the Transformer variant to produce meaningful captions. Without it, the more expressive mapper collapses to near-zero BLEU-4, performing far worse than the simple MLP baseline, which highlights how external grounding is critical when training a high-capacity bridge under a constrained budget.

Comparison against ClipCap and SmallCap shows that CapPix Transformer achieves a competitive METEOR score despite training on substantially less data, suggesting that integrating retrieval into training—rather than only at inference—provides a real semantic grounding advantage. The ablation study over prefix length reveals a meaningful trade-off between caption quality and computational cost: longer prefixes improve METEOR but nearly double trainable parameters and training time, a consideration that matters in Colab-constrained settings.

The main limitations of this work are the small training subset relative to standard benchmarks, the absence of CIDEr and SPICE in evaluation, and the loss of spatial information inherent in global CLIP embeddings. Addressing these through larger-scale training, richer metrics, and region-level visual features remains an important direction for future work. Nonetheless, the methodology and findings reported here offer a reproducible and accessible foundation for further research on lightweight vision-language generation.

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